Survey on State of the Art Research in Speech Recognition

Ines Yang
University of California, Berkeley | Berkeley, CA
inesyang@berkeley.edu

Abstract

Speech recognition is one of the main subfields widely researched in the world of natural language processing. As one of the first areas of research in natural language processing, there have been big strides made in the progress of creating more advanced systems. This paper will introduce some of the current state of the art research and practices in speech recognition, as well as explore a couple of use cases, notably in accent and speech impairment recognition. Throughout this paper, I will introduce the basic components of speech recognition, dive into the current state of the art technologies and their subsequent popular schools of thought, and outline the importance of these techniques in the progress of speech recognition development. [1]

1 Introduction

Speech recognition, which is also referred to as "automatic speech recognition" (ASR) or "speech-to-text" (STT), is a subfield of natural language processing that focuses on enabling computers to understand and interpret human speech. It encapsulates the process of converting spoken language into text form. Speech recognition has come a long way since its birth in the 1960s. What started out as simple research to translate spoken single digits into text, has now expanded into advanced digital assistant systems and powerfully accurate transcription services.

The main ways that speech recognition models detect language is by extracting features that distinguish "sounds" from speech, such as phonemes, which are described as "units of sound", prosody which are hierarchically greater than phonemes and describe the melody or rhythm of language, and other aspects such as background noise. [Jurafsky and Martin, 2024]

Along with these characteristics of speech, speech recognition models have advanced to the point where instead of being given these features to identify language from, neural networks have allowed models to detect speech without the aid of supervision. Because of the powerful pattern-matching capabilities of neural networks, amazingly accurate models have been developed, which will be discussed in further detail in this paper.

[1]Word Count: 2235

2 Current State of the Art Technologies

2.1 How it works

The most recent state of the art research revolving around speech recognition is neural network-based. Within neural networks, there are a large variety of subfields, such as Recurrent Neural Networks (RNN), Convolution Neural Networks (CNN), and Long Short-Term Memory (LSTM), and a hybrid of these techniques. In the past, Hidden Markov Models were the dominant method for speech recognition. They work by representing speech as a sequence of hidden states, and map the probabilities of emitting certain acoustical properties given the previous state (HMM). However, the limitation with this method was that it was too simple since language could not be captured just based on its "previous state" and couldn't capture the complexities of speech.

2.2 Neural Networks

The structure of neural networks mimic the human brain and its neurons, by creating a web of connected nodes and in turn, artificially learning. This type of model is particularly good at recognizing patterns, which is why neural networks are a popular state of the art choice for creating models that need to identify patterns in spoken language. An example of a popular neural network is Wav2Vec, which uses a CNN architecture and implements self-supervised training where it creates its own learning objectives. This model type has been fine tuned to various subtopics such as vq-wav2vec, which utilizes vector quantization to increase the efficiency of its discrete speech representations.[Baevski et al., 2019] Because of the success of neural network models such as wav2vec and vq-wav2vec, many researches have expanded upon this rudimentary idea and have come up with subtypes of neural networks such as recurrent neural networks.

2.2.1 Recurrent Neural Networks

Recurrent Neural Networks are sequential models that are good at processing data over time by using previous words to build up context and understand relationships within a sentence. RNN's have state-of-the-art performance when it comes to tasks such as handwriting recognition, but on its own, does not perform as well when it comes to speech recognition. [Graves et al., 2013] However, a case study showed that when paired with an End-to-End Connectionist Temporal Classification (CTC), its flexibility and long range context created an RNN based model that achieved a test set error of 17.7% which is the best recorded score on the TIMIT phoneme recognition benchmark. [Graves et al., 2013]

2.3 Transformers

Another method has been able to outperform the traditional neural networks. This influential technique is the attention-based transformer, which was first introduced in 2017 in the paper "Attention is All You Need" [Vaswani et al., 2017]. Transformers revolutionized the natural language processing (NLP) field because of its superior performance to RNN's and CNN's in both efficiency and ability to capture long range dependencies. Because of the transformer's ability to parallelize computations, it naturally beats out RNN, which is sequential in nature.

Thus, many newer models use an underlying transformer because of its capabilities to contextualize and capture complex relationships. Specifically in speech recognition, attention based transformers have been used to identify prosody in untranscribed conversational English speech [Roll et al., 2023] ASR has historically struggled with accuracy in conversational speech, so the use of transformers in this area is promising.

2.4 End to End Models

One of the most recent technologies to emerge from speech recognition is End-to-End models which broadly encompass models that take in raw input data and directly outputs the desired output. Already there have been modifications made to E2E models to improve subfields of speech recognition, such as accents. At the AI Institute of X-LANCE, a novel "layer adapter" was added on top of the E2E ASR model encoder which transformed the input acoustic feature into an accent-related feature through the linear combination of all accent bases. [Gong et al., 2021]

2.5 Hybrid Models

With the many of the benefits of different neural network architectures, many speech recognition systems have resorted to combining methods in order to reap the benefits of multiple techniques. These next examples show how hybrid models can outperform models with only one method. Conformers are a prime example of state of the art hybrid models that marries CNN with transformers. [Gulati et al., 2020] This happy marriage consists of CNN, which exploits local features, and transformers which excel at capturing global content well. Using the LibriSpeech benchmark, the model achieves a Word Error Rate (WER) of 2.1% without using a language model and a 1.9% with an external language model. Another model which achieves an accuracy of 72.39%, combines Connectionist Temporal Classification (CTC) and Attention Transformers [Gao et al., 2021]. The model was able to leverage the strengths of CTC which was good at handling variable speech lengths and was tolerant to errors and noise with attention transformers which were great for long range dependencies and parallel processing to achieve greater accuracy and flexibility.

In another case study, an End-to-End model was created for greater customization by incorporating a two-pass system which consisted of an LSTM acoustical model in the first pass, and an E2E model in the second pass [Ye et al., 2021]. This resulted in a 8-10% reduction in word error rate compared to each method's error rate individually. One of the main benefits of this hybrid system is that the first pass does what E2E does poorly at, such as sophisticated speech segmentation and external language model incorporation. By splitting the process into 'passes', the model has an added benefit of the second pass being an "error corrector". On their own, these methods had their weaknesses, but together they achieved greater results.

Another example of utilizing hybrid models for benefits is the Hybrid Transducer and Attention based Encoder-Decoder Model, which traditionally have been used separately [Tang et al., 2023]. This model works by swapping out the predictor in the transducer with the decoder in AED. Some of the benefits from the transducer is maintaining its streaming property while leveraging the powerful contextual advantage that AED brings. This results in a Transducer and Attention based Encoder-Decoder (TAED) model that outperforms transducer for offline ASR and in the streaming case.

2.5.1 Multimodal Models

A subtopic of hybrid models lies in the types of input that ASR takes in. Traditionally, only speech is used to determine speech recognition. However, different types of inputs such as visual learning styles can be incorporated to enhance performance. Using multiple modes of input data is called multimodal modeling. This technique is effective because it pairs together two types of inputs which can inform the other. More multimodal research is being done to strengthen the robustness of the traditional audio modality. [Hu et al., 2023b] For example, if the sound is muffled but the lip readings from the visuals are clear, then a more informed decision can be made. Different methods such as MIR-GAN, focuses on shared dependencies between modalities to achieve this goal. [Hu et al., 2023a] Additionally, there has been work done to enable models that are trained on a single modality to be applicable to multiple modalities [Cheng et al., 2023] and models which utilize cross-modal alignment in pre-training. [Yu et al., 2023]

3 Applications

3.1 Accents

Speech recognition in the past has suffered from not being able to generalize well to non-labeled data. Accents represent one area of speech recognition that has sparse data. To compensate for the lack of data, improvements have been made in this subfield such as using cross-attention with a trainable set of codeblocks [Prabhu et al., 2023] and deep learning neural networks like DeepSpeech to has been used to recognize Indian-English accents.[Dubey and Shah, 2022] Additionally, researchers have identified features such as suprasegmental information which has been beneficial for creating models that are challenged by speech representation disentanglement. [Wang et al., 2023] Self-supervised neural networks have also been employed for both the tasks of accent identification and accented speech recognition. [Deng et al., 2021] There are many accents that are hard to obtain data in, such as African accents which has over 100+ accents. [Owodunni et al., 2024] However, research is being done to group together similar accents to capture nuanced patterns from these accents, and has shown progress mapping in geographic and genealogical similarities. [Owodunni et al., 2024] For other accents, pretrained models still yield high results even if more advanced state of the art techniques aren't used. An example of this is improved voice-controlled systems in Vietnam [Ta et al., 2022] and using regional accented Indian English to fine tune a CNN ASR for indian-english accent. [Chattopadhyay et al., 2022]

3.2 Dysarthria

Dysarthria is another application that has been improved upon in speech recognition. Dysarthria is a motor speech disorder which affects articulation, so speakers with dysarthria have a hard time pronouncing words and effectively communicating. In addition to data scarcity, diversity in the manifestation of speech adds to the complexity of Dysarthria ASR. Aspects of dysarthria, such as severity and speaker-identity are being studied and incorporated into models to produce statistically significant word error rates [Geng et al., 2023]

4 Challenges

Despite the immense progress that has been made in this field, there are still many obstacles to overcome. One of the main challenges that ASR faces is its application to different subfields in NLP. One such example of this is stacking ASR with Named Entity Recognition (NER). Szymanski's paper titled "Why Aren't We NER Yet?" criticizes the performance of NER models when being directly fed the raw ASR output. [Szymański et al., 2023] The paper talks about how ASR from spontaneous speech, which is characterized by human speech which usually lacks formal grammar structure, is especially difficult to perform NER on.

One of the next steps forward in the development of ASR seems to be in relation to the other functions that "stack" on top of it. Common "stacking" functions include NER as discussed, and more popularly, speech translation which includes tasks such as language translation and sentiment analysis.

5 Improvements

In light of the challenges that speech recognition models face, they have also seen great improvements in End-to-End modeling and in specific areas such as accent recognition. One of the main areas that can improve a model's performance is to exploit context clues. If a model is given prior information about potentially relevant words or phrases, then it will have a much better change of predicting correctly. This technique is known as contextual biasing. In one paper, they improved upon contextual biasing in E2E modeling by adding a "spelling correction model" on top of their E2E. [Wang et al., 2022] This method acts as a sort of preprocessing filtering step. This extra step proves to be very effective, with an observed word error rate reduction of 51% over traditional biasing methods, and speeds up inference by 2.1 times.

Another novel improvement is focused on the consistency of performance across different accents. In the past, speech recognition for accents have been solved by creating a fine tuned accent-specific ASR, which commonly uses labeled trained data. [Kothawade et al., 2023] However, this model is not very time or cost effective because of the large amounts of data required for each accent-specific model. Thus, Data-efficient and faIr Targeted subseT selectiOn (DITTO) was created to find the most informative set of data that represented the target accent within a fixed budget. This model expands upon supervised learning models by creating a more efficient way to create labeled data. Compared to other speech selection methods such as Indic-TTS and L2, DITTO is 3-5 times as label-efficient.

Conclusion

Speech recognition has come a long way since its birth. With state of the art research and work done in deep learning techniques such as neural networks, attention-based models, and hybrid architectures, this subfield has become more accurate and efficient. With these newfound advancements however, there are still many challenges. The application of ASR to other tasks in NLP still remains something to be improved upon. Despite these challenges, great strides are being made in this field to improve model performance. With these improvements, speech recognition has the potential to enhance the user experience of human to machine for all people, regardless of accent or speech impairment.

Acknowledgements

I would like to acknowledge Behrang Mohit, the professor of INFO 159, and the staff of INFO 159 for guidance in writing this paper.

References

Alexei Baevski, Steffen Schneider, and Michael Auli. vq-wav2vec: Self-supervised learning of discrete speech representations. *CoRR*, abs/1910.05453, 2019. URL http://arxiv.org/abs/1910.05453.

Sourav Chattopadhyay, Junaid Hamid Bhat, Sowmya Rasipuram, Aditi Debsharma, and Anutosh Maitra. Improving performance of nemo asr system for indian accent english. In *2022 IEEE 19th India Council International Conference (INDICON)*, pages 1–5, 2022. doi: 10.1109/INDICON56171.2022.10039815. URL https://ieeexplore.ieee.org/document/10039815.

Xize Cheng, Tao Jin, Linjun Li, Wang Lin, Xinyu Duan, and Zhou Zhao. OpenSR: Open-modality speech recognition via maintaining multi-modality alignment. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6592–6607, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.363. URL https://aclanthology.org/2023.acl-long.363.

Keqi Deng, Songjun Cao, and Long Ma. Improving accent identification and accented speech recognition under a framework of self-supervised learning, 2021. URL https://arxiv.org/pdf/2109.07349.

Priyank Dubey and Bilal Shah. Deep speech based end-to-end automated speech recognition (asr) for indian-english accents, 2022. URL https://arxiv.org/abs/2204.00977.

Qiang Gao, Haiwei Wu, Yanqing Sun, and Yitao Duan. An end-to-end speech accent recognition method based on hybrid ctc/attention transformer asr. In *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 7253–7257, 2021. doi: 10.1109/ICASSP39728.2021.9414082. URL https://arxiv.org/abs/2110.04891.

Mengzhe Geng, Zengrui Jin, Tianzi Wang, Shujie Hu, Jiajun Deng, Mingyu Cui, Guinan Li, Jianwei Yu, Xurong Xie, and Xunying Liu. Use of speech impairment severity for dysarthric speech recognition, 2023. URL https://arxiv.org/abs/2305.10659.

Xun Gong, Yizhou Lu, Zhikai Zhou, and Yanmin Qian. Layer-wise fast adaptation for end-to-end multi-accent speech recognition. In *Interspeech 2021*, interspeech$_2021.ISCA, August2021.$ *doi* : . URL http://dx.doi.org/10.21437/Interspeech.2021-1075.

Alex Graves, Abdel-rahman Mohamed, and Geoffrey E. Hinton. Speech recognition with deep recurrent neural networks. *CoRR*, abs/1303.5778, 2013. URL http://arxiv.org/abs/1303.5778.

Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, and Ruoming Pang. Conformer: Convolution-augmented transformer for speech recognition, 2020. URL https://arxiv.org/abs/2005.08100.

Yuchen Hu, Chen Chen, Ruizhe Li, Heqing Zou, and Eng Siong Chng. MIR-GAN: Refining frame-level modality-invariant representations with adversarial network for audio-visual speech recognition. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11610–11625, Toronto, Canada, July 2023a. Association for Computational Linguistics. 10.18653/v1/2023.acl-long.649. URL https://aclanthology.org/2023.acl-long.649.

Yuchen Hu, Ruizhe Li, Chen Chen, Chengwei Qin, Qiu-Shi Zhu, and Eng Siong Chng. Hearing lips in noise: Universal viseme-phoneme mapping and transfer for robust audio-visual speech recognition. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15213–15232, Toronto, Canada, July 2023b. Association for Computational Linguistics. 10.18653/v1/2023.acl-long.848. URL https://aclanthology.org/2023.acl-long.848.

Daniel Jurafsky and James H. Martin. *Speech and Language Processing*. 2024. URL web.stanford.edu/~jurafsky/slp3/. Draft.

Suraj Kothawade, Anmol Mekala, D.Chandra Sekhara Hetha Havya, Mayank Kothyari, Rishabh Iyer, Ganesh Ramakrishnan, and Preethi Jyothi. DITTO: Data-efficient and fair targeted subset selection for ASR accent adaptation. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5810–5822, Toronto, Canada, July 2023. Association for Computational Linguistics. 10.18653/v1/2023.acl-long.319. URL https://aclanthology.org/2023.acl-long.319.

Abraham Owodunni, Aditya Yadavalli, Chris Emezue, Tobi Olatunji, and Clinton Mbataku. AccentFold: A journey through African accents for zero-shot ASR adaptation to target accents. In Yvette Graham and Matthew Purver, editors, *Findings of the Association for Computational Linguistics: EACL 2024*, pages 2146–2161, St. Julian's, Malta, March 2024. Association for Computational Linguistics. URL https://aclanthology.org/2024.findings-eacl.142.

Darshan Prabhu, Preethi Jyothi, Sriram Ganapathy, and Vinit Unni. Accented speech recognition with accent-specific codebooks. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7175–7188, Singapore, December 2023. Association for Computational Linguistics. 10.18653/v1/2023.emnlp-main.444. URL https://aclanthology.org/2023.emnlp-main.444.

Nathan Roll, Calbert Graham, and Simon Todd. PSST! prosodic speech segmentation with transformers. In Jing Jiang, David Reitter, and Shumin Deng, editors, *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*, pages 476–487, Singapore, December 2023. Association for Computational Linguistics. 10.18653/v1/2023.conll-1.31. URL https://aclanthology.org/2023.conll-1.31.

Piotr Szymański, Lukasz Augustyniak, Mikolaj Morzy, Adrian Szymczak, Krzysztof Surdyk, and Piotr Żelasko. Why aren't we NER yet? artifacts of ASR errors in named entity recognition in spontaneous speech transcripts. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1746–1761, Toronto, Canada, July 2023. Association for Computational Linguistics. 10.18653/v1/2023.acl-long.98. URL https://aclanthology.org/2023.acl-long.98.

Bao Thang Ta, Xuan Vuong Dang, Quang Tien Duong, Nhat Minh Le, and Van Hai Do. Improving vietnamese accent recognition using asr transfer learning. In *2022 25th Conference of the Oriental COCOSDA International Committee* *for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA)*, pages 1–6, 2022. 10.1109/O-COCOSDA202257103.2022.9997947. URL https://ieeexplore.ieee.org/document/9997947.

Yun Tang, Anna Sun, Hirofumi Inaguma, Xinyue Chen, Ning Dong, Xutai Ma, Paden Tomasello, and Juan Pino. Hybrid transducer and attention based encoder-decoder modeling for speech-to-text tasks. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12441–12455, Toronto, Canada, July 2023. Association for Computational Linguistics. 10.18653/v1/2023.acl-long.695. URL https://aclanthology.org/2023.acl-long.695.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. *CoRR*, abs/1706.03762, 2017. URL http://arxiv.org/abs/1706.03762.

Linqin Wang, Zhengtao Yu, Yuanzhang Yang, Shengxiang Gao, Cunli Mao, and Yuxin Huang. Non-parallel accent transfer based on fine-grained controllable accent modelling. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9288–9298, Singapore, December 2023. Association for Computational Linguistics. 10.18653/v1/2023.findings-emnlp.622. URL https://aclanthology.org/2023.findings-emnlp.622.

Xiaoqiang Wang, Yanqing Liu, Jinyu Li, Veljko Miljanic, Sheng Zhao, and Hosam Khalil. Towards contextual spelling correction for customization of end-to-end speech recognition systems, 2022. URL https://arxiv.org/abs/2203.00888.

Guoli Ye, Vadim Mazalov, Jinyu Li, and Yifan Gong. Have best of both worlds: two-pass hybrid and E2E cascading framework for speech recognition. *CoRR*, abs/2110.04891, 2021. URL https://arxiv.org/abs/2110.04891.

Tianshu Yu, Haoyu Gao, Ting-En Lin, Min Yang, Yuchuan Wu, Wentao Ma, Chao Wang, Fei Huang, and Yongbin Li. Speech-text pre-training for spoken dialog understanding with explicit cross-modal alignment. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7900–7913, Toronto, Canada, July 2023. Association for Computational Linguistics. 10.18653/v1/2023.acl-long.438. URL https://aclanthology.org/2023.acl-long.438.